MODULE 12 · DOMAIN 3 - APPLICATION DEVELOPMENT

Model hubs, model cards and experiment metrics

Reading the label to build a shortlist, then letting your own experiments pick the winner.

SECTION 3 of 6 EXAM WEIGHT 30% MODULE 12 of 40

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A model card is a nutrition label. **Read it to build the shortlist. Then let your own experiments choose the winner.**

Where models come from

SOURCE	WHAT YOU GET
system.ai in Unity Catalog	Pre-installed foundation models, permissive licences, ready to serve
Databricks Marketplace	Model families you install into Unity Catalog
Hugging Face Hub	Download a model and register it in Unity Catalog
Foundation Model APIs	Databricks-hosted models, called by endpoint name
External providers	Models hosted elsewhere, governed through Databricks

ON THE EXAM

Whatever the source, the model ends up governed in Unity Catalog. Models in system.ai are available to all account users by default.

What a model card tells you

SECTION	WHAT TO LOOK FOR
Intended use	The tasks it was built and tuned for
Training data	What it learned from, and the cut-off date
Languages	Which languages it actually supports
Size and context	Parameters, architecture, context window
Evaluation	Benchmark results, and on which tasks
Limitations	Known weaknesses, biases, unsafe uses
Licence	Whether your intended use is permitted

ON THE EXAM

Licence, languages and intended use can each disqualify a model on their own. Check them before any benchmark.

Signals on a hub listing, and how much to trust them

- **Task tags** and **licence tags**: fast, reliable filters.
- **Variant:** base, instruct or chat. It changes how the model behaves.
- **Size, languages and last updated**: practical fit and maintenance.
- **Downloads and likes** show popularity, not fitness for your task.
- A missing or empty card is itself a warning sign.

> ON THE EXAM
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> Popularity is not quality. The most downloaded model may be English-only, or licensed for research use only.

Base or instruct?

BASE MODELS

Trained to continue text

- No instruction tuning
- Rambles or continues the prompt
- The starting point for fine-tuning
- Not for serving a chat assistant

INSTRUCT OR CHAT MODELS

Tuned to follow instructions

- Answers the question asked
- Respects system prompts and formats
- The right choice for serving
- Can still be fine-tuned further

THE TIEBREAKER

Databricks recommends base versions for fine-tuning and instruct versions for deployment and serving.

The card's claims, or your experiment?

THE CARD'S BENCHMARKS

General ability

- Public datasets
- Someone else's tasks
- Useful for a first shortlist
- Cannot see your documents

YOUR EXPERIMENT

Fitness for your job

- Your own evaluation set
- Your prompts and retrieval
- The right metric for the task
- Picks from the shortlist

THE TIEBREAKER

Card to shortlist, experiments to decide. When they disagree, your experiment wins.

The right metric for the task

TASK	COMMON METRIC	BETTER IS
Classification	Accuracy, and F1 for imbalanced classes	Higher
Summarisation	ROUGE, overlap with reference summaries	Higher
Translation	BLEU, overlap with reference translations	Higher
Extraction	Exact match or field-level accuracy	Higher
Open answers, rag	LLM judges: correctness, groundedness	Higher pass rate
Language modelling	Perplexity	Lower
Operational	Latency, cost per request, toxicity	Lower

ON THE EXAM

Match the metric to the task first. A strong score on the wrong metric tells you nothing.

Higher is better, or lower is better?

HIGHER IS BETTER

Quality scores

- Accuracy and F1
- ROUGE and BLEU
- Exact match
- Judge pass rates

LOWER IS BETTER

Errors and costs

- Perplexity
- Latency
- Cost per request
- Toxicity

THE TIEBREAKER

Perplexity is how surprised the model is by the text. Less surprise is better. Picking the highest perplexity is a classic wrong answer.

When a good-looking number misleads

- **Accuracy with imbalanced classes:** predicting the majority class scores well and finds nothing.
- **BLEU and ROUGE** reward word overlap, not correctness. A fluent wrong answer can score well.
- **Perplexity** measures fluency, not whether the task was done.
- One metric alone hides trade-offs. Always read quality with latency and cost.
- Compare runs only on the **same evaluation set**.

> ON THE EXAM
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> When classes are imbalanced, prefer F1, precision and recall over accuracy.

Three runs, one requirement

RUN	ROUGE-L	P95 LATENCY	RELATIVE COST
Small model	0.38	0.7 s	1x
Mid-size model	0.43	1.3 s	2x
Frontier model	0.45	3.4 s	8x
Requirement	at least 0.42	under 2 s	lowest that passes

ON THE EXAM

Apply the thresholds first: the small model fails quality, the frontier model fails latency. The mid-size model is the only one that passes.

> A company needs to summarise customer emails written in Japanese, for a commercial product.
>
> The engineer is comparing model listings on a hub.

Which model should go on the shortlist?

A One whose card lists Japanese, summarisation and commercial use

B The most downloaded summarisation model, which is English-only

C The top benchmark scorer, licensed for non-commercial use only

D The most recently uploaded base model in the summarisation list

◆ PAUSE AND DECIDE

Which model should go on the shortlist?

A Japanese, summarisation, commercial

Passes all three disqualifiers: language, task, licence.

B Most downloaded, English-only

Popular, but cannot handle the language.

C Top benchmark, non-commercial

The licence forbids the intended use.

D Newest base model

Recency is not fit, and base models are for fine-tuning.

THE SIGNAL

Japanese, summarise, commercial product. Three requirements, three card sections: languages, intended use, licence.

A team serves a customer chat assistant using a model from system.ai. It chose the base version of the model family.

Responses ramble, ignore the system prompt, and sometimes just continue the user's message.

What should the team change?

A Use a larger base model from the same family of models

B Raise the temperature so replies vary a little more

C Serve the instruct version; keep base for fine-tuning

D Add more retrieved context to each prompt it receives

◆ PAUSE AND DECIDE

What should the team change?

A **Larger base model**
Still not instruction-tuned. Rambles at greater length.

B **Higher temperature**
More randomness, less control.

C **Instruct version**
Tuned to follow instructions. Databricks' own recommendation.

D **More context**
More text for a base model to continue.

THE SIGNAL

Ignores the system prompt and continues the user's message. That is how a base model behaves.

> A bank compares two models for flagging suspicious transactions. Only 2% of transactions in the evaluation set are suspicious.
>
> Model X: accuracy 98.1%, F1 0.12. Model Y: accuracy 96.5%, F1 0.71.

Which model should the bank choose?

A. Model X, because its accuracy is the higher of the two

B. Model Y, because its F1 reflects the rare suspicious cases

C. Whichever model has the lower perplexity on the test set

D. Neither: compare the two models on their BLEU scores instead

◆ PAUSE AND DECIDE

Which model should the bank choose?

A Model X, higher accuracy

98% is roughly what always saying not suspicious would score.

B Model Y, higher F1

F1 measures success on the rare class that matters.

C Lower perplexity

Fluency metric. Irrelevant to classification.

D BLEU

A translation metric. Wrong task.

THE SIGNAL

Only 2% are suspicious. With imbalanced classes, accuracy flatters. F1 tells the truth about the rare class.

> An extraction task requires at least 95% field accuracy and a p95 latency under 1.5 seconds, at the lowest cost that meets both.
>
> Runs: small model 93%, 0.4 s. Mid-size model 96.2%, 1.2 s. Large model 97.0%, 3.8 s. Mid-size reasoning model 96.8%, 2.6 s.

Which run should the team choose?

A The large model, which has the highest accuracy

B The small model, which is fastest and cheapest

C The mid-size reasoning model, the second most accurate

D The mid-size model, the third most accurate

◆ PAUSE AND DECIDE

Which run should the team choose?

A Large model

3.8 s fails the latency requirement.

B Small model

93% fails the accuracy requirement.

C Mid-size reasoning model

2.6 s fails the latency requirement.

D Mid-size model

96.2% and 1.2 s: the only run passing both.

THE SIGNAL

Apply every threshold first, then optimise what remains. The best score and the lowest cost both failed a requirement.

What to carry forward

- Sources: **system.ai**, Marketplace, Hugging Face, all governed in Unity Catalog.
- Read the card for **licence, languages and intended use** first. Popularity is not fit.
- **Instruct** to serve. **Base** to fine-tune.
- Match metric to task. F1 for imbalance. Lower perplexity, latency and cost are better.
- Thresholds first, then the cheapest run that passes.

> ON THE EXAM
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> Next: module 13 chooses an embedding model, and its context length, for your documents and queries.